NLP Assignment 0

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4.

a.

We claim that the output of the model is always greater or equal 0.5, Therefore every sample will get a positive label and for the given dataset the maximal accuracy will be $\frac{1}{5}$.

$$p\left(\text{paraphrase}|x_1, x_2\right) = \sigma\left(\text{relu}\left(x_1\right)^\top \text{relu}\left(x_1\right)\right) = \sigma\left(\sum_{k=1}^{d} \max\left(0, x_{1,k}\right) \cdot \max\left(0, x_{2,k}\right)\right)$$
$$\overset{*}{\geq} \sigma\left(\sum_{k=1}^{d} 0 \cdot 0\right) = 0.5$$

* is derived from σ monotonicity and sum of non-negative parts.

b.

By removing the relu function we can get a negative input to σ and therefore a negative result. We will choose the new classifier:

$$p\left(\text{paraphrase}|x_1, x_2\right) = \sigma\left(x_2^\top x_1\right)$$

c.

I will go over each choice and explain why i chose/ignored it.

-Accuracy, is actually bad in our situation, For example a classifier which always predicts false will achieve 80%.

-Precision, is greatly affected from the imbalance in the dataset, because it uses the negative samples for normalization.

-Recall, is a good choice because it is not affected by the imbalance in the dataset, it only compares sizes within the positive samples.

-ROC-AUC, is excellent choice for evaluating the model ability to separate classes while not being affected from the imbalance.

-AUC-PR, is using precision as one of it's axis, from the same reasons we ignore it.

-Confusion Matrix, is also a good choice as it gives us a general idea of the model performance on the dataset but we should not forget the imbalance when interpreting it.

AI Disclosure

We have used Chat-GPT for understanding Google Colab platform, some ML basic ideas recap and for explaining different metrics for evaluating a model (also some Youtube videos about ROC).

We have also used Gemini inside Google Colab notebook for some help on the numpy and matplotlib syntax.

We used the AI in a moderate way which allowed us to boost productivity while still learning effectively.